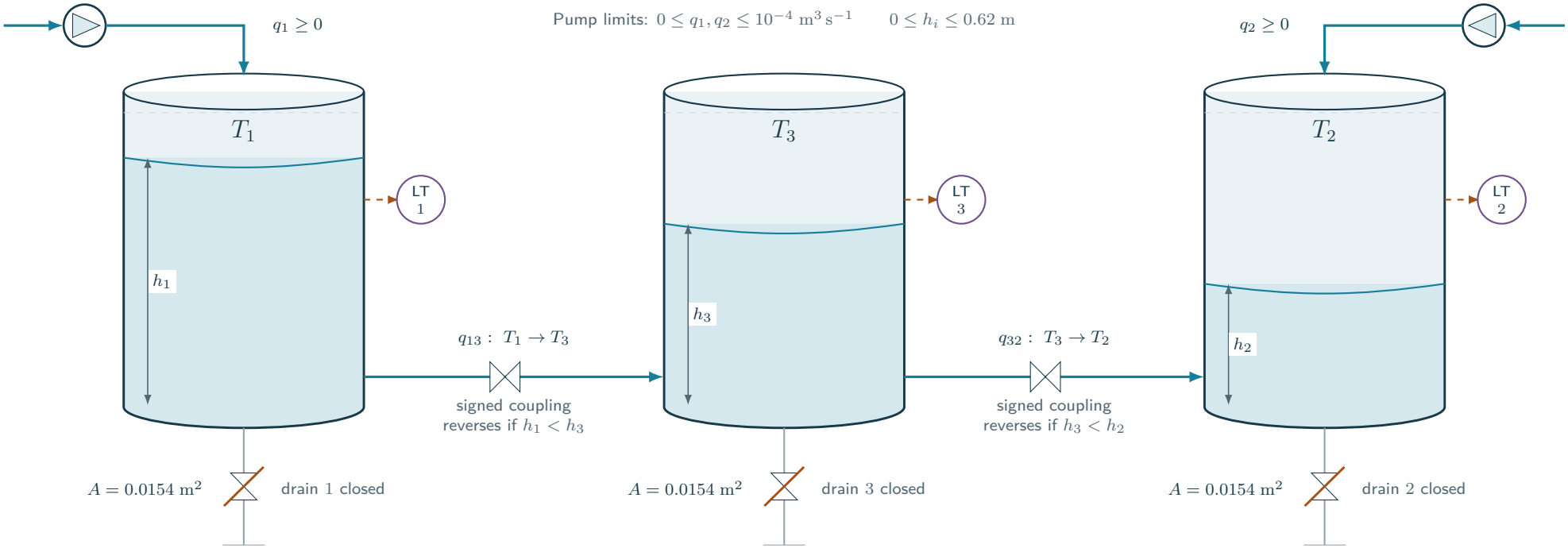
Three interconnected tanks

Paper section 3.7: *Three-tank-system*

Physical arrangement: T_1–T_3–T_2 | State ordering in the implementation: $[h_1, h_2, h_3]^\mathsf{T}$



Blue: liquid flows. Dashed orange: level observations. Crossed valves: zero nominal discharge; drain/fault experiments are explicit extensions.

Hydraulic coupling

$$q_{ij} = a\,\mu\,\mathrm{sgn}(h_i - h_j) \cdot \sqrt{2g|h_i - h_j|}$$

$a = 0.5\times10^{-4}\ \mathrm{m^2}$, $\mu = 0.48$.

Nominal level balances

$$A\dot{h}_1 = q_1 - q_{13}$$
$$A\dot{h}_3 = q_{13} - q_{32}$$
$$A\dot{h}_2 = q_2 + q_{32}$$

Time in seconds; levels in metres.

Conservation and operating limit

$$\frac{d}{dt}A(h_1 + h_2 + h_3) = q_1 + q_2.$$

With closed drains, positive pumping rules out a steady equilibrium. Corrected area unit: $\mathrm{m^2}$.

Scope: three-state nominal model, Eq. (36). Vessel heights in this drawing are illustrative. A tracking experiment with continuous pumping requires an explicitly declared outlet extension.
Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.